

DSA Mastery Sheet

1000 problems grouped by data structure and pattern, ordered Easy > Medium > Hard

1000 problems | 205 Easy | 591 Medium | 204 Hard

How to read this

LC # is the LeetCode problem number - search it in the LeetCode app or type it into the problem-set search box. A * after the number means LeetCode Premium is needed. **GFG** means a classic problem that is not free on LeetCode (search the name on GeeksforGeeks). Tick the box when solved. Inside each pattern the order is the study order: finish the group before moving on. Do a first pass on Easy + Medium, then come back for Hard and anything you could not solve cold.

Contents

Arrays and Hashing (59 problems)	3
Two Pointers (34 problems)	5
Sliding Window (39 problems)	7
Prefix Sum and Intervals (41 problems)	9
Binary Search (50 problems)	11
Strings (43 problems)	13
Matrix (24 problems)	15
Linked List (45 problems)	16
Stack and Monotonic Stack (54 problems)	18
Queue Deque and Design (24 problems)	20
Recursion and Backtracking (48 problems)	21
Binary Trees (80 problems)	23
Binary Search Tree (25 problems)	26
Heap and Priority Queue (41 problems)	27
Trie (14 problems)	29
Graphs (107 problems)	30
Dynamic Programming (144 problems)	34
Greedy (47 problems)	39
Bit Manipulation (28 problems)	41
Math and Number Theory (33 problems)	42
Sorting and Advanced DS (20 problems)	43

Quick Pattern Checklist 44

Arrays and Hashing (59 problems)

Ti ck	Sr	LC #	Problem	Level
Hash Map - Lookup and Complement - 13 problems				
<i>Idea: Store what you've seen; look up (target - x) in O(1)</i>				
☐	1	1	Two Sum	Easy
☐	2	217	Contains Duplicate	Easy
☐	3	219	Contains Duplicate II	Easy
☐	4	383	Ransom Note	Easy
☐	5	242	Valid Anagram	Easy
☐	6	349	Intersection of Two Arrays	Easy
☐	7	350	Intersection of Two Arrays II	Easy
☐	8	202	Happy Number	Easy
☐	9	205	Isomorphic Strings	Easy
☐	10	290	Word Pattern	Easy
☐	11	1679	Max Number of K-Sum Pairs	Medium
☐	12	554	Brick Wall	Medium
☐	13	454	4Sum II	Medium
Hash Map - Frequency and Grouping - 18 problems				
<i>Idea: Count with a map, or group by a canonical key (sorted string / count tuple)</i>				
☐	14	1512	Number of Good Pairs	Easy
☐	15	1189	Maximum Number of Balloons	Easy
☐	16	387	First Unique Character in a String	Easy
☐	17	169	Majority Element	Easy
☐	18	136	Single Number	Easy
☐	19	1160	Find Words That Can Be Formed by Characters	Easy
☐	20	705	Design HashSet	Easy
☐	21	706	Design HashMap	Easy
☐	22	49	Group Anagrams	Medium
☐	23	347	Top K Frequent Elements	Medium
☐	24	451	Sort Characters By Frequency	Medium
☐	25	791	Custom Sort String	Medium
☐	26	1347	Minimum Number of Steps to Make Two Strings Anagram	Medium
☐	27	36	Valid Sudoku	Medium
☐	28	609	Find Duplicate File in System	Medium
☐	29	380	Insert Delete GetRandom O(1)	Medium
☐	30	229	Majority Element II	Medium
☐	31	128	Longest Consecutive Sequence	Medium

In-place Array Manipulation - 22 problems

Idea: Read/write pointers, overwrite instead of allocating, reverse-three-times tricks

☐	32	27	Remove Element	Easy
☐	33	26	Remove Duplicates from Sorted Array	Easy
☐	34	283	Move Zeroes	Easy
☐	35	88	Merge Sorted Array	Easy
☐	36	66	Plus One	Easy
☐	37	561	Array Partition	Easy
☐	38	414	Third Maximum Number	Easy
☐	39	118	Pascal's Triangle	Easy
☐	40	119	Pascal's Triangle II	Easy
☐	41	1752	Check if Array Is Sorted and Rotated	Easy
☐	42	121	Best Time to Buy and Sell Stock	Easy
☐	43	628	Maximum Product of Three Numbers	Easy
☐	44	1588	Sum of All Odd Length Subarrays	Easy
☐	45	122	Best Time to Buy and Sell Stock II	Medium
☐	46	189	Rotate Array	Medium
☐	47	80	Remove Duplicates from Sorted Array II	Medium
☐	48	238	Product of Array Except Self	Medium
☐	49	31	Next Permutation	Medium
☐	50	75	Sort Colors	Medium
☐	51	334	Increasing Triplet Subsequence	Medium
☐	52	2149	Rearrange Array Elements by Sign	Medium
☐	53	324	Wiggle Sort II	Medium

Cyclic Sort / Index as Hash - 6 problems

Idea: Values in 1..n -> put v at index v-1, or mark presence by negating

☐	54	268	Missing Number	Easy
☐	55	448	Find All Numbers Disappeared in an Array	Easy
☐	56	645	Set Mismatch	Easy
☐	57	442	Find All Duplicates in an Array	Medium
☐	58	287	Find the Duplicate Number	Medium
☐	59	41	First Missing Positive	Hard

Two Pointers (34 problems)

Ti ck	Sr	LC #	Problem	Level
Opposite Ends on Sorted / Symmetric Data - 17 problems				
<i>Idea: Move the pointer whose move can still improve the answer</i>				
☐	60	125	Valid Palindrome	Easy
☐	61	344	Reverse String	Easy
☐	62	345	Reverse Vowels of a String	Easy
☐	63	680	Valid Palindrome II	Easy
☐	64	977	Squares of a Sorted Array	Easy
☐	65	905	Sort Array By Parity	Easy
☐	66	167	Two Sum II - Input Array Is Sorted	Medium
☐	67	11	Container With Most Water	Medium
☐	68	15	3Sum	Medium
☐	69	16	3Sum Closest	Medium
☐	70	611	Valid Triangle Number	Medium
☐	71	633	Sum of Square Numbers	Medium
☐	72	881	Boats to Save People	Medium
☐	73	1877	Minimize Maximum Pair Sum in Array	Medium
☐	74	151	Reverse Words in a String	Medium
☐	75	18	4Sum	Medium
☐	76	42	Trapping Rain Water	Hard
Same Direction / Subsequence Matching - 8 problems				
<i>Idea: Slow pointer writes or matches, fast pointer scans</i>				
☐	77	392	Is Subsequence	Easy
☐	78	844	Backspace String Compare	Easy
☐	79	925	Long Pressed Name	Easy
☐	80	1332	Remove Palindromic Subsequences	Easy
☐	81	165	Compare Version Numbers	Medium
☐	82	1750	Minimum Length of String After Deleting Similar Ends	Medium
☐	83	581	Shortest Unsorted Continuous Subarray	Medium
☐	84	845	Longest Mountain in Array	Medium
Merge and Intersect Sorted Inputs - 8 problems				
<i>Idea: Advance the pointer sitting on the smaller value</i>				
☐	85	1385	Find the Distance Value Between Two Arrays	Easy
☐	86	2824	Count Pairs Whose Sum is Less than Target	Easy
☐	87	455	Assign Cookies	Easy
☐	88	986	Interval List Intersections	Medium
☐	89	658	Find K Closest Elements	Medium
☐	90	1577	Number of Ways Where Square of Number Is Equal to Product of Two Numbers	Medium
☐	91	948	Bag of Tokens	Medium
☐	92	1010	Pairs of Songs With Total Durations Divisible by 60	Medium

Fast and Slow on Arrays - 1 problems

Idea: Treat index -> value as a next pointer; Floyd cycle detection



93

457

Circular Array Loop

Medium

Sliding Window (39 problems)

Ti ck	Sr	LC #	Problem	Level
Fixed-Size Window - 12 problems				
<i>Idea: Add the new right element, drop the old left element, update the answer</i>				
☐	94	643	Maximum Average Subarray I	Easy
☐	95	1652	Defuse the Bomb	Easy
☐	96	1456	Maximum Number of Vowels in a Substring of Given Length	Medium
☐	97	1343	Number of Sub-arrays of Size K and Average Greater than or Equal to Threshold	Medium
☐	98	1052	Grumpy Bookstore Owner	Medium
☐	99	1423	Maximum Points You Can Obtain from Cards	Medium
☐	100	438	Find All Anagrams in a String	Medium
☐	101	567	Permutation in String	Medium
☐	102	187	Repeated DNA Sequences	Medium
☐	103	2461	Maximum Sum of Distinct Subarrays With Length K	Medium
☐	104	239	Sliding Window Maximum	Hard
☐	105	30	Substring with Concatenation of All Words	Hard
Variable Window - Longest Valid - 14 problems				
<i>Idea: Expand right; shrink left while invalid; record max length</i>				
☐	106	3	Longest Substring Without Repeating Characters	Medium
☐	107	424	Longest Repeating Character Replacement	Medium
☐	108	1004	Max Consecutive Ones III	Medium
☐	109	1493	Longest Subarray of 1's After Deleting One Element	Medium
☐	110	904	Fruit Into Baskets	Medium
☐	111	GFG	Longest Substring with At Most K Distinct Characters	Medium
☐	112	1208	Get Equal Substrings Within Budget	Medium
☐	113	1838	Frequency of the Most Frequent Element	Medium
☐	114	1695	Maximum Erasure Value	Medium
☐	115	2401	Longest Nice Subarray	Medium
☐	116	2516	Take K of Each Character From Left and Right	Medium
☐	117	1438	Longest Continuous Subarray With Absolute Diff Less Than or Equal to Limit	Medium
☐	118	978	Longest Turbulent Subarray	Medium
☐	119	395	Longest Substring with At Least K Repeating Characters	Medium

Variable Window - Shortest / Counting - 13 problems*Idea: Shrink while valid for shortest; count = atMost(k) - atMost(k-1)*

☐	120	209	Minimum Size Subarray Sum	Medium
☐	121	713	Subarray Product Less Than K	Medium
☐	122	1248	Count Number of Nice Subarrays	Medium
☐	123	930	Binary Subarrays With Sum	Medium
☐	124	1358	Number of Substrings Containing All Three Characters	Medium
☐	125	1658	Minimum Operations to Reduce X to Zero	Medium
☐	126	1234	Replace the Substring for Balanced String	Medium
☐	127	2962	Count Subarrays Where Max Element Appears at Least K Times	Medium
☐	128	992	Subarrays with K Different Integers	Hard
☐	129	2444	Count Subarrays With Fixed Bounds	Hard
☐	130	76	Minimum Window Substring	Hard
☐	131	GFG	Minimum Window Subsequence	Hard
☐	132	862	Shortest Subarray with Sum at Least K	Hard

Prefix Sum and Intervals (41 problems)

Tick	Sr	LC #	Problem	Level
Prefix Sum - 10 problems				
<i>Idea: $sum(i..j) = P[j+1] - P[i]$; precompute once, answer queries in $O(1)$</i>				
☐	133	1480	Running Sum of 1d Array	Easy
☐	134	303	Range Sum Query - Immutable	Easy
☐	135	724	Find Pivot Index	Easy
☐	136	1732	Find the Highest Altitude	Easy
☐	137	1413	Minimum Value to Get Positive Step by Step Sum	Easy
☐	138	2270	Number of Ways to Split Array	Medium
☐	139	304	Range Sum Query 2D - Immutable	Medium
☐	140	1109	Corporate Flight Bookings	Medium
☐	141	1094	Car Pooling	Medium
☐	142	GFG	Range Addition	Medium
Prefix Sum + Hash Map - 7 problems				
<i>Idea: Count subarrays by storing how many times each prefix value has occurred</i>				
☐	143	560	Subarray Sum Equals K	Medium
☐	144	523	Continuous Subarray Sum	Medium
☐	145	974	Subarray Sums Divisible by K	Medium
☐	146	525	Contiguous Array	Medium
☐	147	1590	Make Sum Divisible by P	Medium
☐	148	GFG	Maximum Size Subarray Sum Equals k	Medium
☐	149	1526	Minimum Number of Increments on Subarrays to Form a Target Array	Hard
Kadane / Max Subarray - 6 problems				
<i>Idea: Best sum ending here = $\max(x, \text{prev} + x)$</i>				
☐	150	53	Maximum Subarray	Medium
☐	151	152	Maximum Product Subarray	Medium
☐	152	918	Maximum Sum Circular Subarray	Medium
☐	153	1749	Maximum Absolute Sum of Any Subarray	Medium
☐	154	1186	Maximum Subarray Sum with One Deletion	Medium
☐	155	1031	Maximum Sum of Two Non-Overlapping Subarrays	Medium
Merge Intervals - 9 problems				
<i>Idea: Sort by start; merge when $\text{next.start} \leq \text{current.end}$</i>				
☐	156	228	Summary Ranges	Easy
☐	157	GFG	Meeting Rooms	Easy
☐	158	56	Merge Intervals	Medium
☐	159	57	Insert Interval	Medium
☐	160	1288	Remove Covered Intervals	Medium
☐	161	729	My Calendar I	Medium
☐	162	731	My Calendar II	Medium
☐	163	352	Data Stream as Disjoint Intervals	Hard
☐	164	GFG	Employee Free Time	Hard

Interval Scheduling and Sweep Line - 9 problems

Idea: Greedy by end time, or sweep over +1/-1 start/end events

☐	165	435	Non-overlapping Intervals	Medium
☐	166	452	Minimum Number of Arrows to Burst Balloons	Medium
☐	167	GFG	Meeting Rooms II	Medium
☐	168	1024	Video Stitching	Medium
☐	169	3169	Count Days Without Meetings	Medium
☐	170	2406	Divide Intervals Into Minimum Number of Groups	Medium
☐	171	1353	Maximum Number of Events That Can Be Attended	Medium
☐	172	732	My Calendar III	Hard
☐	173	1851	Minimum Interval to Include Each Query	Hard

Binary Search (50 problems)

Ti ck	Sr	LC #	Problem	Level
Classic Binary Search - 12 problems				
<i>Idea: Halve the search space each step; be strict about lo/hi and loop invariant</i>				
☐	174	704	Binary Search	Easy
☐	175	35	Search Insert Position	Easy
☐	176	278	First Bad Version	Easy
☐	177	374	Guess Number Higher or Lower	Easy
☐	178	69	Sqrt(x)	Easy
☐	179	367	Valid Perfect Square	Easy
☐	180	441	Arranging Coins	Easy
☐	181	744	Find Smallest Letter Greater Than Target	Easy
☐	182	1539	Kth Missing Positive Number	Easy
☐	183	1608	Special Array With X Elements Greater Than or Equal X	Easy
☐	184	34	Find First and Last Position of Element in Sorted Array	Medium
☐	185	540	Single Element in a Sorted Array	Medium
Peak / Mountain - 3 problems				
<i>Idea: Compare mid with its neighbour to decide which side is rising</i>				
☐	186	852	Peak Index in a Mountain Array	Medium
☐	187	162	Find Peak Element	Medium
☐	188	1095	Find in Mountain Array	Hard
Rotated Sorted Array - 4 problems				
<i>Idea: One half is always sorted; check which half holds the target</i>				
☐	189	33	Search in Rotated Sorted Array	Medium
☐	190	153	Find Minimum in Rotated Sorted Array	Medium
☐	191	81	Search in Rotated Sorted Array II	Medium
☐	192	154	Find Minimum in Rotated Sorted Array II	Hard
Search in 2D / Sorted Matrix - 4 problems				
<i>Idea: Flatten to 1D indices, or walk the staircase from a corner</i>				
☐	193	1351	Count Negative Numbers in a Sorted Matrix	Easy
☐	194	74	Search a 2D Matrix	Medium
☐	195	240	Search a 2D Matrix II	Medium
☐	196	378	Kth Smallest Element in a Sorted Matrix	Medium

Binary Search on Answer - 20 problems

Idea: Monotone feasible(x): binary search the smallest/largest x that works

☐	197	875	Koko Eating Bananas	Medium
☐	198	1011	Capacity To Ship Packages Within D Days	Medium
☐	199	1283	Find the Smallest Divisor Given a Threshold	Medium
☐	200	1482	Minimum Number of Days to Make m Bouquets	Medium
☐	201	1552	Magnetic Force Between Two Balls	Medium
☐	202	2187	Minimum Time to Complete Trips	Medium
☐	203	1870	Minimum Speed to Arrive on Time	Medium
☐	204	2064	Minimized Maximum of Products Distributed to Any Store	Medium
☐	205	2226	Maximum Candies Allocated to K Children	Medium
☐	206	2616	Minimize the Maximum Difference of Pairs	Medium
☐	207	1802	Maximum Value at a Given Index in a Bounded Array	Medium
☐	208	1201	Ugly Number III	Medium
☐	209	GFG	Aggressive Cows	Medium
☐	210	GFG	Allocate Minimum Number of Pages	Medium
☐	211	410	Split Array Largest Sum	Hard
☐	212	GFG	Painter's Partition Problem	Hard
☐	213	878	Nth Magical Number	Hard
☐	214	668	Kth Smallest Number in Multiplication Table	Hard
☐	215	719	Find K-th Smallest Pair Distance	Hard
☐	216	2141	Maximum Running Time of N Computers	Hard

Binary Search + Sorted Structure - 7 problems

Idea: Precompute an order, then lower/upper-bound each query

☐	217	2300	Successful Pairs of Spells and Potions	Medium
☐	218	475	Heaters	Medium
☐	219	275	H-Index II	Medium
☐	220	436	Find Right Interval	Medium
☐	221	981	Time Based Key-Value Store	Medium
☐	222	528	Random Pick with Weight	Medium
☐	223	4	Median of Two Sorted Arrays	Hard

Strings (43 problems)

Tick	Sr	LC #	Problem	Level
Basic String Handling - 29 problems				
<i>Idea: Iterate carefully; think in terms of char codes, carry, and edge cases</i>				
☐	224	58	Length of Last Word	Easy
☐	225	14	Longest Common Prefix	Easy
☐	226	541	Reverse String II	Easy
☐	227	557	Reverse Words in a String III	Easy
☐	228	520	Detect Capital	Easy
☐	229	434	Number of Segments in a String	Easy
☐	230	824	Goat Latin	Easy
☐	231	482	License Key Formatting	Easy
☐	232	13	Roman to Integer	Easy
☐	233	171	Excel Sheet Column Number	Easy
☐	234	168	Excel Sheet Column Title	Easy
☐	235	67	Add Binary	Easy
☐	236	415	Add Strings	Easy
☐	237	1662	Check If Two String Arrays are Equivalent	Easy
☐	238	1768	Merge Strings Alternately	Easy
☐	239	917	Reverse Only Letters	Easy
☐	240	443	String Compression	Medium
☐	241	38	Count and Say	Medium
☐	242	12	Integer to Roman	Medium
☐	243	6	Zigzag Conversion	Medium
☐	244	43	Multiply Strings	Medium
☐	245	8	String to Integer (atoi)	Medium
☐	246	179	Largest Number	Medium
☐	247	1657	Determine if Two Strings Are Close	Medium
☐	248	271*	Encode and Decode Strings	Medium
☐	249	833	Find and Replace in String	Medium
☐	250	2131	Longest Palindrome by Concatenating Two Letter Words	Medium
☐	251	68	Text Justification	Hard
☐	252	65	Valid Number	Hard
Palindromes - 6 problems				
<i>Idea: Expand around center, or DP on (i, j); Manacher for O(n)</i>				
☐	253	9	Palindrome Number	Easy
☐	254	1328	Break a Palindrome	Medium
☐	255	5	Longest Palindromic Substring	Medium
☐	256	647	Palindromic Substrings	Medium
☐	257	214	Shortest Palindrome	Hard
☐	258	336	Palindrome Pairs	Hard

Pattern Matching (KMP / Z / Rolling Hash) - 8 problems

Idea: Reuse matched prefix information instead of restarting

☐	259	28	Find the Index of the First Occurrence in a String	Easy
☐	260	459	Repeated Substring Pattern	Easy
☐	261	796	Rotate String	Easy
☐	262	686	Repeated String Match	Medium
☐	263	1392	Longest Happy Prefix	Hard
☐	264	2223	Sum of Scores of Built Strings	Hard
☐	265	1044	Longest Duplicate Substring	Hard
☐	266	1316	Distinct Echo Substrings	Hard

Matrix (24 problems)

Tick	Sr	LC #	Problem	Level
Matrix Traversal and Simulation - 24 problems <i>Idea: Track direction and boundaries; layer-by-layer</i>				
☐	267	867	Transpose Matrix	Easy
☐	268	832	Flipping an Image	Easy
☐	269	566	Reshape the Matrix	Easy
☐	270	766	Toeplitz Matrix	Easy
☐	271	1380	Lucky Numbers in a Matrix	Easy
☐	272	1572	Matrix Diagonal Sum	Easy
☐	273	1672	Richest Customer Wealth	Easy
☐	274	661	Image Smoother	Easy
☐	275	1252	Cells with Odd Values in a Matrix	Easy
☐	276	1260	Shift 2D Grid	Easy
☐	277	2133	Check if Every Row and Column Contains All Numbers	Easy
☐	278	1886	Determine Whether Matrix Can Be Obtained By Rotation	Easy
☐	279	54	Spiral Matrix	Medium
☐	280	59	Spiral Matrix II	Medium
☐	281	885	Spiral Matrix III	Medium
☐	282	48	Rotate Image	Medium
☐	283	73	Set Matrix Zeroes	Medium
☐	284	498	Diagonal Traverse	Medium
☐	285	1329	Sort the Matrix Diagonally	Medium
☐	286	289	Game of Life	Medium
☐	287	1706	Where Will the Ball Fall	Medium
☐	288	1901	Find a Peak Element II	Medium
☐	289	1284	Minimum Number of Flips to Convert Binary Matrix to Zero Matrix	Hard
☐	290	1632	Rank Transform of a Matrix	Hard

Linked List (45 problems)

Ti ck	Sr	LC #	Problem	Level
Cycle Detection Chain - 5 problems				
<i>Idea: Floyd's slow/fast pointers - detect, then length, then start, then remove</i>				
☐	291	141	Linked List Cycle	Easy
☐	292	GFG	Length of Loop in Linked List	Easy
☐	293	160	Intersection of Two Linked Lists	Easy
☐	294	142	Linked List Cycle II	Medium
☐	295	GFG	Remove Loop in Linked List	Medium
Middle and Halving (Fast/Slow) - 6 problems				
<i>Idea: Fast moves 2x; slow lands on middle / n-th from end</i>				
☐	296	876	Middle of the Linked List	Easy
☐	297	234	Palindrome Linked List	Easy
☐	298	19	Remove Nth Node From End of List	Medium
☐	299	2095	Delete the Middle Node of a Linked List	Medium
☐	300	143	Reorder List	Medium
☐	301	2130	Maximum Twin Sum of a Linked List	Medium
Reversal Variants - 7 problems				
<i>Idea: Three-pointer reversal (prev, curr, next); use a dummy head for sub-ranges</i>				
☐	302	206	Reverse Linked List	Easy
☐	303	92	Reverse Linked List II	Medium
☐	304	24	Swap Nodes in Pairs	Medium
☐	305	1721	Swapping Nodes in a Linked List	Medium
☐	306	61	Rotate List	Medium
☐	307	GFG	Reverse Alternating K Nodes	Medium
☐	308	25	Reverse Nodes in k-Group	Hard
Basic Node Operations - 12 problems				
<i>Idea: Dummy head + careful pointer rewiring</i>				
☐	309	1290	Convert Binary Number in a Linked List to Integer	Easy
☐	310	203	Remove Linked List Elements	Easy
☐	311	83	Remove Duplicates from Sorted List	Easy
☐	312	GFG	Delete N Nodes After M Nodes of a Linked List	Easy
☐	313	237	Delete Node in a Linked List	Medium
☐	314	328	Odd Even Linked List	Medium
☐	315	82	Remove Duplicates from Sorted List II	Medium
☐	316	725	Split Linked List in Parts	Medium
☐	317	1669	Merge In Between Linked Lists	Medium
☐	318	2487	Remove Nodes From Linked List	Medium
☐	319	1171	Remove Zero Sum Consecutive Nodes from Linked List	Medium
☐	320	2058	Find the Minimum and Maximum Number of Nodes Between Critical Points	Medium

Merge, Sort and Partition - 5 problems

Idea: Merge two sorted lists is the building block for everything here

☐	321	21	Merge Two Sorted Lists	Easy
☐	322	86	Partition List	Medium
☐	323	147	Insertion Sort List	Medium
☐	324	148	Sort List	Medium
☐	325	23	Merge k Sorted Lists	Hard

Arithmetic on Lists - 5 problems

Idea: Simulate digit-by-digit with carry

☐	326	2181	Merge Nodes in Between Zeros	Medium
☐	327	2	Add Two Numbers	Medium
☐	328	445	Add Two Numbers II	Medium
☐	329	2816	Double a Number Represented as a Linked List	Medium
☐	330	GFG	Plus One Linked List	Medium

Cloning and Multi-level Lists - 5 problems

Idea: Interleave / map old->new nodes; recurse for child pointers

☐	331	138	Copy List with Random Pointer	Medium
☐	332	430	Flatten a Multilevel Doubly Linked List	Medium
☐	333	707	Design Linked List	Medium
☐	334	146	LRU Cache	Medium
☐	335	460	LFU Cache	Hard

Stack and Monotonic Stack (54 problems)

Tick	Sr	LC #	Problem	Level
Matching Brackets - 13 problems				
<i>Idea: Push openers; pop and compare on closers; leftover = invalid</i>				
☐	336	20	Valid Parentheses	Easy
☐	337	1021	Remove Outermost Parentheses	Easy
☐	338	1544	Make The String Great	Easy
☐	339	1047	Remove All Adjacent Duplicates In String	Easy
☐	340	682	Baseball Game	Easy
☐	341	1209	Remove All Adjacent Duplicates in String II	Medium
☐	342	921	Minimum Add to Make Parentheses Valid	Medium
☐	343	1249	Minimum Remove to Make Valid Parentheses	Medium
☐	344	1963	Minimum Number of Swaps to Make the String Balanced	Medium
☐	345	856	Score of Parentheses	Medium
☐	346	678	Valid Parenthesis String	Medium
☐	347	1190	Reverse Substrings Between Each Pair of Parentheses	Medium
☐	348	32	Longest Valid Parentheses	Hard
Expression Evaluation and Parsing - 9 problems				
<i>Idea: Operand stack + operator stack, or recursion per parenthesis</i>				
☐	349	150	Evaluate Reverse Polish Notation	Medium
☐	350	227	Basic Calculator II	Medium
☐	351	394	Decode String	Medium
☐	352	71	Simplify Path	Medium
☐	353	735	Asteroid Collision	Medium
☐	354	636	Exclusive Time of Functions	Medium
☐	355	224	Basic Calculator	Hard
☐	356	726	Number of Atoms	Hard
☐	357	1106	Parsing A Boolean Expression	Hard

Monotonic Stack - Next Greater / Smaller - 15 problems*Idea: Keep the stack sorted; a pop means you found the answer for that element*

☐	358	496	Next Greater Element I	Easy
☐	359	1475	Final Prices With a Special Discount in a Shop	Easy
☐	360	503	Next Greater Element II	Medium
☐	361	739	Daily Temperatures	Medium
☐	362	901	Online Stock Span	Medium
☐	363	1019	Next Greater Node In Linked List	Medium
☐	364	853	Car Fleet	Medium
☐	365	402	Remove K Digits	Medium
☐	366	316	Remove Duplicate Letters	Medium
☐	367	1081	Smallest Subsequence of Distinct Characters	Medium
☐	368	907	Sum of Subarray Minimums	Medium
☐	369	2104	Sum of Subarray Ranges	Medium
☐	370	456	132 Pattern	Medium
☐	371	962	Maximum Width Ramp	Medium
☐	372	2289	Steps to Make Array Non-decreasing	Medium

Histogram and Rectangle Problems - 6 problems*Idea: For each bar, find nearest smaller on both sides*

☐	373	1856	Maximum Subarray Min-Product	Medium
☐	374	1504	Count Submatrices With All Ones	Medium
☐	375	84	Largest Rectangle in Histogram	Hard
☐	376	85	Maximal Rectangle	Hard
☐	377	1793	Maximum Score of a Good Subarray	Hard
☐	378	1944	Number of Visible People in a Queue	Hard

Stack Design and Simulation - 11 problems*Idea: Combine stacks/queues to expose the required interface*

☐	379	225	Implement Stack using Queues	Easy
☐	380	232	Implement Queue using Stacks	Easy
☐	381	155	Min Stack	Medium
☐	382	1381	Design a Stack With Increment Operation	Medium
☐	383	1472	Design Browser History	Medium
☐	384	946	Validate Stack Sequences	Medium
☐	385	341	Flatten Nested List Iterator	Medium
☐	386	385	Mini Parser	Medium
☐	387	331	Verify Preorder Serialization of a Binary Tree	Medium
☐	388	895	Maximum Frequency Stack	Hard
☐	389	1172	Dinner Plate Stacks	Hard

Queue Deque and Design (24 problems)

Ti ck	Sr	LC #	Problem	Level
Queue Basics and Simulation - 10 problems				
<i>Idea: FIFO simulation; circular buffers for fixed capacity</i>				
☐	390	933	Number of Recent Calls	Easy
☐	391	2073	Time Needed to Buy Tickets	Easy
☐	392	1700	Number of Students Unable to Eat Lunch	Easy
☐	393	1603	Design Parking System	Easy
☐	394	622	Design Circular Queue	Medium
☐	395	641	Design Circular Deque	Medium
☐	396	950	Reveal Cards In Increasing Order	Medium
☐	397	649	Dota2 Senate	Medium
☐	398	1352	Product of the Last K Numbers	Medium
☐	399	1670	Design Front Middle Back Queue	Medium
Monotonic Deque - 3 problems				
<i>Idea: Deque holds candidates in sorted order; evict from front when out of window</i>				
☐	400	1696	Jump Game VI	Medium
☐	401	1425	Constrained Subsequence Sum	Hard
☐	402	1499	Max Value of Equation	Hard
System-style Design - 11 problems				
<i>Idea: Pick hash map + list/heap combos to hit $O(1)/O(\log n)$ per op</i>				
☐	403	1166*	Design File System	Medium
☐	404	1146	Snapshot Array	Medium
☐	405	1797	Design Authentication Manager	Medium
☐	406	1396	Design Underground System	Medium
☐	407	355	Design Twitter	Medium
☐	408	353*	Design Snake Game	Medium
☐	409	348*	Design Tic-Tac-Toe	Medium
☐	410	GFG	Design Hit Counter	Medium
☐	411	855	Exam Room	Medium
☐	412	642*	Design Search Autocomplete System	Hard
☐	413	432	All O`one Data Structure	Hard

Recursion and Backtracking (48 problems)

Tick	Sr	LC #	Problem	Level
Recursion Warm-up - 4 problems				
<i>Idea: Define base case, trust the smaller call, combine results</i>				
☐	414	509	Fibonacci Number	Easy
☐	415	231	Power of Two	Easy
☐	416	50	Pow(x, n)	Medium
☐	417	GFG	Tower of Hanoi	Medium
Subsets and Combinations - 12 problems				
<i>Idea: Choose / skip each element; sort + skip duplicates at the same depth</i>				
☐	418	78	Subsets	Medium
☐	419	90	Subsets II	Medium
☐	420	77	Combinations	Medium
☐	421	39	Combination Sum	Medium
☐	422	40	Combination Sum II	Medium
☐	423	216	Combination Sum III	Medium
☐	424	17	Letter Combinations of a Phone Number	Medium
☐	425	22	Generate Parentheses	Medium
☐	426	1286	Iterator for Combination	Medium
☐	427	526	Beautiful Arrangement	Medium
☐	428	2597	The Number of Beautiful Subsets	Medium
☐	429	491	Non-decreasing Subsequences	Medium
Permutations - 7 problems				
<i>Idea: Track used[] or swap in place; skip duplicate values at the same level</i>				
☐	430	784	Letter Case Permutation	Medium
☐	431	46	Permutations	Medium
☐	432	47	Permutations II	Medium
☐	433	1980	Find Unique Binary String	Medium
☐	434	1415	The k-th Lexicographical String of All Happy Strings of Length n	Medium
☐	435	60	Permutation Sequence	Hard
☐	436	996	Number of Squareful Arrays	Hard
Partitioning and String Backtracking - 7 problems				
<i>Idea: Try every cut position; prune when the prefix is invalid</i>				
☐	437	93	Restore IP Addresses	Medium
☐	438	131	Palindrome Partitioning	Medium
☐	439	306	Additive Number	Medium
☐	440	1593	Split a String Into the Max Number of Unique Substrings	Medium
☐	441	140	Word Break II	Hard
☐	442	282	Expression Add Operators	Hard
☐	443	301	Remove Invalid Parentheses	Hard

Grid and Constraint Backtracking - 18 problems

Idea: Mark visited, recurse to neighbours, un-mark; prune early

☐	444	79	Word Search	Medium
☐	445	1219	Path with Maximum Gold	Medium
☐	446	GFG	Rat in a Maze	Medium
☐	447	473	Matchsticks to Square	Medium
☐	448	698	Partition to K Equal Sum Subsets	Medium
☐	449	1239	Maximum Length of a Concatenated String with Unique Characters	Medium
☐	450	967	Numbers With Same Consecutive Differences	Medium
☐	451	89	Gray Code	Medium
☐	452	357	Count Numbers with Unique Digits	Medium
☐	453	2305	Fair Distribution of Cookies	Medium
☐	454	1718	Construct the Lexicographically Largest Valid Sequence	Medium
☐	455	980	Unique Paths III	Hard
☐	456	51	N-Queens	Hard
☐	457	52	N-Queens II	Hard
☐	458	37	Sudoku Solver	Hard
☐	459	1255	Maximum Score Words Formed by Letters	Hard
☐	460	1307	Verbal Arithmetic Puzzle	Hard
☐	461	679	24 Game	Hard

Binary Trees (80 problems)

Tick	Sr	LC #	Problem	Level
DFS Traversals and Recursion Basics - 21 problems <i>Idea: Solve for left, solve for right, combine at the node</i>				
☐	462	94	Binary Tree Inorder Traversal	Easy
☐	463	144	Binary Tree Preorder Traversal	Easy
☐	464	145	Binary Tree Postorder Traversal	Easy
☐	465	104	Maximum Depth of Binary Tree	Easy
☐	466	111	Minimum Depth of Binary Tree	Easy
☐	467	100	Same Tree	Easy
☐	468	101	Symmetric Tree	Easy
☐	469	226	Invert Binary Tree	Easy
☐	470	572	Subtree of Another Tree	Easy
☐	471	110	Balanced Binary Tree	Easy
☐	472	543	Diameter of Binary Tree	Easy
☐	473	617	Merge Two Binary Trees	Easy
☐	474	965	Univalued Binary Tree	Easy
☐	475	872	Leaf-Similar Trees	Easy
☐	476	563	Binary Tree Tilt	Easy
☐	477	404	Sum of Left Leaves	Easy
☐	478	589	N-ary Tree Preorder Traversal	Easy
☐	479	559	Maximum Depth of N-ary Tree	Easy
☐	480	951	Flip Equivalent Binary Trees	Medium
☐	481	623	Add One Row to Tree	Medium
☐	482	1367	Linked List in Binary Tree	Medium

BFS and Level Order - 16 problems*Idea: Queue + process one level at a time using the queue size*

☐	483	637	Average of Levels in Binary Tree	Easy
☐	484	993	Cousins in Binary Tree	Easy
☐	485	102	Binary Tree Level Order Traversal	Medium
☐	486	107	Binary Tree Level Order Traversal II	Medium
☐	487	103	Binary Tree Zigzag Level Order Traversal	Medium
☐	488	199	Binary Tree Right Side View	Medium
☐	489	429	N-ary Tree Level Order Traversal	Medium
☐	490	515	Find Largest Value in Each Tree Row	Medium
☐	491	116	Populating Next Right Pointers in Each Node	Medium
☐	492	117	Populating Next Right Pointers in Each Node II	Medium
☐	493	958	Check Completeness of a Binary Tree	Medium
☐	494	1302	Deepest Leaves Sum	Medium
☐	495	1609	Even Odd Tree	Medium
☐	496	1161	Maximum Level Sum of a Binary Tree	Medium
☐	497	513	Find Bottom Left Tree Value	Medium
☐	498	662	Maximum Width of Binary Tree	Medium

Root-to-Leaf and Path Problems - 14 problems*Idea: Pass running state down; use a global for best 'bent' path*

☐	499	112	Path Sum	Easy
☐	500	257	Binary Tree Paths	Easy
☐	501	1022	Sum of Root To Leaf Binary Numbers	Easy
☐	502	113	Path Sum II	Medium
☐	503	437	Path Sum III	Medium
☐	504	129	Sum Root to Leaf Numbers	Medium
☐	505	1448	Count Good Nodes in Binary Tree	Medium
☐	506	1457	Pseudo-Palindromic Paths in a Binary Tree	Medium
☐	507	988	Smallest String Starting From Leaf	Medium
☐	508	687	Longest Univalue Path	Medium
☐	509	1372	Longest ZigZag Path in a Binary Tree	Medium
☐	510	GFG	Binary Tree Longest Consecutive Sequence	Medium
☐	511	1080	Insufficient Nodes in Root to Leaf Paths	Medium
☐	512	124	Binary Tree Maximum Path Sum	Hard

Construction and Serialization - 7 problems*Idea: Root from preorder/postorder head; split the inorder array around it*

☐	513	606	Construct String from Binary Tree	Easy
☐	514	105	Construct Binary Tree from Preorder and Inorder Traversal	Medium
☐	515	106	Construct Binary Tree from Inorder and Postorder Traversal	Medium
☐	516	889	Construct Binary Tree from Preorder and Postorder Traversal	Medium
☐	517	654	Maximum Binary Tree	Medium
☐	518	114	Flatten Binary Tree to Linked List	Medium
☐	519	297	Serialize and Deserialize Binary Tree	Hard

LCA and Distance Between Nodes - 7 problems

Idea: If both sides return non-null, the current node is the LCA

☐	520	236	Lowest Common Ancestor of a Binary Tree	Medium
☐	521	1123	Lowest Common Ancestor of Deepest Leaves	Medium
☐	522	1110	Delete Nodes And Return Forest	Medium
☐	523	2096	Step-By-Step Directions From a Binary Tree Node to Another	Medium
☐	524	863	All Nodes Distance K in Binary Tree	Medium
☐	525	2385	Amount of Time for Binary Tree to Be Infected	Medium
☐	526	1361	Validate Binary Tree Nodes	Medium

Tree DP (Return Info from Children) - 11 problems

Idea: Post-order returns a tuple; combine at parent

☐	527	508	Most Frequent Subtree Sum	Medium
☐	528	2265	Count Nodes Equal to Average of Subtree	Medium
☐	529	1026	Maximum Difference Between Node and Ancestor	Medium
☐	530	979	Distribute Coins in Binary Tree	Medium
☐	531	652	Find Duplicate Subtrees	Medium
☐	532	337	House Robber III	Medium
☐	533	222	Count Complete Tree Nodes	Medium
☐	534	1339	Maximum Product of Splitted Binary Tree	Medium
☐	535	1145	Binary Tree Coloring Game	Medium
☐	536	968	Binary Tree Cameras	Hard
☐	537	834	Sum of Distances in Tree	Hard

Views and Column Order - 4 problems

Idea: BFS/DFS with (row, col) coordinates; group by column

☐	538	GFG	Binary Tree Vertical Order Traversal	Medium
☐	539	GFG	Top View of Binary Tree	Medium
☐	540	GFG	Boundary Traversal of Binary Tree	Medium
☐	541	987	Vertical Order Traversal of a Binary Tree	Hard

Binary Search Tree (25 problems)

Ti ck	Sr	LC #	Problem	Level
BST Search / Insert / Delete - 6 problems				
<i>Idea: Go left if smaller, right if larger; delete via successor</i>				
☐	542	700	Search in a Binary Search Tree	Easy
☐	543	270*	Closest Binary Search Tree Value	Easy
☐	544	701	Insert into a Binary Search Tree	Medium
☐	545	450	Delete Node in a BST	Medium
☐	546	235	Lowest Common Ancestor of a Binary Search Tree	Medium
☐	547	285*	Inorder Successor in BST	Medium
Inorder = Sorted - 9 problems				
<i>Idea: Inorder traversal of a BST yields ascending order</i>				
☐	548	530	Minimum Absolute Difference in BST	Easy
☐	549	501	Find Mode in Binary Search Tree	Easy
☐	550	653	Two Sum IV - Input is a BST	Easy
☐	551	938	Range Sum of BST	Easy
☐	552	230	Kth Smallest Element in a BST	Medium
☐	553	98	Validate Binary Search Tree	Medium
☐	554	173	Binary Search Tree Iterator	Medium
☐	555	538	Convert BST to Greater Tree	Medium
☐	556	99	Recover Binary Search Tree	Medium
Build and Rebalance - 10 problems				
<i>Idea: Pick the middle as root for balance; use preorder + bounds to rebuild</i>				
☐	557	108	Convert Sorted Array to Binary Search Tree	Easy
☐	558	109	Convert Sorted List to Binary Search Tree	Medium
☐	559	1382	Balance a Binary Search Tree	Medium
☐	560	669	Trim a Binary Search Tree	Medium
☐	561	1008	Construct Binary Search Tree from Preorder Traversal	Medium
☐	562	449	Serialize and Deserialize BST	Medium
☐	563	96	Unique Binary Search Trees	Medium
☐	564	95	Unique Binary Search Trees II	Medium
☐	565	333*	Largest BST Subtree	Medium
☐	566	220	Contains Duplicate III	Hard

Heap and Priority Queue (41 problems)

Ti ck	Sr	LC #	Problem	Level
Top-K Elements - 16 problems				
<i>Idea: Keep a heap of size k; the root is the k-th best</i>				
☐	567	703	Kth Largest Element in a Stream	Easy
☐	568	1046	Last Stone Weight	Easy
☐	569	506	Relative Ranks	Easy
☐	570	1337	The K Weakest Rows in a Matrix	Easy
☐	571	215	Kth Largest Element in an Array	Medium
☐	572	973	K Closest Points to Origin	Medium
☐	573	692	Top K Frequent Words	Medium
☐	574	373	Find K Pairs with Smallest Sums	Medium
☐	575	786	Kth Smallest Prime Fraction	Medium
☐	576	1481	Least Number of Unique Integers after K Removals	Medium
☐	577	1338	Reduce Array Size to The Half	Medium
☐	578	1962	Remove Stones to Minimize the Total	Medium
☐	579	2530	Maximal Score After Applying K Operations	Medium
☐	580	2583	Kth Largest Sum in a Binary Tree	Medium
☐	581	1054	Distant Barcodes	Medium
☐	582	1705	Maximum Number of Eaten Apples	Medium
Two Heaps - 4 problems				
<i>Idea: Max-heap for the lower half, min-heap for the upper half</i>				
☐	583	295	Find Median from Data Stream	Hard
☐	584	480	Sliding Window Median	Hard
☐	585	502	IPO	Hard
☐	586	1675	Minimize Deviation in Array	Hard

Greedy with Heap (Scheduling) - 18 problems

Idea: Always take the best available option, refill as time moves

☐	587	GFG	Minimum Cost to Connect Sticks	Medium
☐	588	621	Task Scheduler	Medium
☐	589	767	Reorganize String	Medium
☐	590	1405	Longest Happy String	Medium
☐	591	1642	Furthest Building You Can Reach	Medium
☐	592	1845	Seat Reservation Manager	Medium
☐	593	1834	Single-Threaded CPU	Medium
☐	594	1882	Process Tasks Using Servers	Medium
☐	595	2462	Total Cost to Hire K Workers	Medium
☐	596	2034	Stock Price Fluctuation	Medium
☐	597	264	Ugly Number II	Medium
☐	598	313	Super Ugly Number	Medium
☐	599	GFG	Rearrange String k Distance Apart	Hard
☐	600	630	Course Schedule III	Hard
☐	601	871	Minimum Number of Refueling Stops	Hard
☐	602	1354	Construct Target Array With Multiple Sums	Hard
☐	603	1383	Maximum Performance of a Team	Hard
☐	604	857	Minimum Cost to Hire K Workers	Hard

Merge K Sorted Streams - 3 problems

Idea: Heap holds one candidate per stream

☐	605	1439	Find the Kth Smallest Sum of a Matrix With Sorted Rows	Hard
☐	606	632	Smallest Range Covering Elements from K Lists	Hard
☐	607	407	Trapping Rain Water II	Hard

Trie (14 problems)

Tick	Sr	LC #	Problem	Level
Trie Basics - 7 problems <i>Idea: Node = children map + end flag; walk one char per step</i>				
☐	608	GFG	Index Pairs of a String	Easy
☐	609	208	Implement Trie (Prefix Tree)	Medium
☐	610	211	Design Add and Search Words Data Structure	Medium
☐	611	648	Replace Words	Medium
☐	612	677	Map Sum Pairs	Medium
☐	613	720	Longest Word in Dictionary	Medium
☐	614	1268	Search Suggestions System	Medium
Trie + Backtracking / Bits - 7 problems <i>Idea: Prune the DFS with the trie; bit-trie for max XOR</i>				
☐	615	421	Maximum XOR of Two Numbers in an Array	Medium
☐	616	212	Word Search II	Hard
☐	617	1032	Stream of Characters	Hard
☐	618	745	Prefix and Suffix Search	Hard
☐	619	425*	Word Squares	Hard
☐	620	472	Concatenated Words	Hard
☐	621	2416	Sum of Prefix Scores of Strings	Hard

Graphs (107 problems)

Ti ck	Sr	LC #	Problem	Level
Graph Basics and Traversal - 14 problems				
<i>Idea: Build adjacency lists first; DFS/BFS with a visited set</i>				
☐	622	1971	Find if Path Exists in Graph	Easy
☐	623	997	Find the Town Judge	Easy
☐	624	1791	Find Center of Star Graph	Easy
☐	625	133	Clone Graph	Medium
☐	626	797	All Paths From Source to Target	Medium
☐	627	1557	Minimum Number of Vertices to Reach All Nodes	Medium
☐	628	1466	Reorder Routes to Make All Paths Lead to the City Zero	Medium
☐	629	1376	Time Needed to Inform All Employees	Medium
☐	630	1311	Get Watched Videos by Your Friends	Medium
☐	631	GFG	Find the Celebrity	Medium
☐	632	841	Keys and Rooms	Medium
☐	633	2101	Detonate the Maximum Bombs	Medium
☐	634	2359	Find Closest Node to Given Two Nodes	Medium
☐	635	2685	Count the Number of Complete Components	Medium
Grid DFS - Flood Fill and Islands - 11 problems				
<i>Idea: Treat cells as nodes; sink visited land; count DFS launches</i>				
☐	636	733	Flood Fill	Easy
☐	637	463	Island Perimeter	Easy
☐	638	200	Number of Islands	Medium
☐	639	695	Max Area of Island	Medium
☐	640	130	Surrounded Regions	Medium
☐	641	417	Pacific Atlantic Water Flow	Medium
☐	642	1254	Number of Closed Islands	Medium
☐	643	1020	Number of Enclaves	Medium
☐	644	1905	Count Sub Islands	Medium
☐	645	GFG	Number of Distinct Islands	Medium
☐	646	827	Making A Large Island	Hard

Multi-source and Grid BFS - 10 problems

Idea: Push all sources first; distance = BFS level

☐	647	994	Rotting Oranges	Medium
☐	648	542	01 Matrix	Medium
☐	649	GFG	Walls and Gates	Medium
☐	650	1162	As Far from Land as Possible	Medium
☐	651	1091	Shortest Path in Binary Matrix	Medium
☐	652	1926	Nearest Exit from Entrance in Maze	Medium
☐	653	934	Shortest Bridge	Medium
☐	654	GFG	Minimum Knight Moves	Medium
☐	655	1293	Shortest Path in a Grid with Obstacles Elimination	Hard
☐	656	1568	Minimum Number of Days to Disconnect Island	Hard

BFS on Implicit State Graphs - 11 problems

Idea: State = node; neighbours = one legal move; BFS gives fewest steps

☐	657	433	Minimum Genetic Mutation	Medium
☐	658	752	Open the Lock	Medium
☐	659	1306	Jump Game III	Medium
☐	660	909	Snakes and Ladders	Medium
☐	661	1654	Minimum Jumps to Reach Home	Medium
☐	662	127	Word Ladder	Hard
☐	663	126	Word Ladder II	Hard
☐	664	815	Bus Routes	Hard
☐	665	773	Sliding Puzzle	Hard
☐	666	864	Shortest Path to Get All Keys	Hard
☐	667	847	Shortest Path Visiting All Nodes	Hard

Topological Sort and Cycle Detection - 17 problems

Idea: Kahn's algorithm (in-degree) or DFS colours

☐	668	GFG	Detect Cycle in an Undirected Graph	Medium
☐	669	GFG	Detect Cycle in a Directed Graph	Medium
☐	670	785	Is Graph Bipartite?	Medium
☐	671	886	Possible Bipartition	Medium
☐	672	207	Course Schedule	Medium
☐	673	210	Course Schedule II	Medium
☐	674	802	Find Eventual Safe States	Medium
☐	675	310	Minimum Height Trees	Medium
☐	676	GFG	Parallel Courses	Medium
☐	677	444*	Sequence Reconstruction	Medium
☐	678	2192	All Ancestors of a Node in a Directed Acyclic Graph	Medium
☐	679	2115	Find All Possible Recipes from Given Supplies	Medium
☐	680	1462	Course Schedule IV	Medium
☐	681	269*	Alien Dictionary	Hard
☐	682	1203	Sort Items by Groups Respecting Dependencies	Hard
☐	683	329	Longest Increasing Path in a Matrix	Hard
☐	684	1857	Largest Color Value in a Directed Graph	Hard

Union Find (DSU) - 17 problems

Idea: Find with path compression, union by rank; count components

☐	685	547	Number of Provinces	Medium
☐	686	GFG	Number of Connected Components in an Undirected Graph	Medium
☐	687	GFG	Graph Valid Tree	Medium
☐	688	684	Redundant Connection	Medium
☐	689	721	Accounts Merge	Medium
☐	690	990	Satisfiability of Equality Equations	Medium
☐	691	947	Most Stones Removed with Same Row or Column	Medium
☐	692	1319	Number of Operations to Make Network Connected	Medium
☐	693	1202	Smallest String With Swaps	Medium
☐	694	399	Evaluate Division	Medium
☐	695	959	Regions Cut by Slashes	Medium
☐	696	685	Redundant Connection II	Hard
☐	697	778	Swim in Rising Water	Hard
☐	698	GFG	Number of Islands II	Hard
☐	699	2092	Find All People With Secret	Hard
☐	700	1579	Remove Max Number of Edges to Keep Graph Fully Traversable	Hard
☐	701	GFG	Optimize Water Distribution in a Village	Hard

Shortest Path - Dijkstra / Bellman-Ford / Floyd - 15 problems

Idea: Non-negative weights: heap; K edges: Bellman-Ford; all-pairs: Floyd

☐	702	GFG	Dijkstra's Algorithm	Medium
☐	703	GFG	Bellman-Ford Algorithm	Medium
☐	704	GFG	Floyd-Warshall Algorithm	Medium
☐	705	743	Network Delay Time	Medium
☐	706	787	Cheapest Flights Within K Stops	Medium
☐	707	1514	Path with Maximum Probability	Medium
☐	708	1631	Path With Minimum Effort	Medium
☐	709	1976	Number of Ways to Arrive at Destination	Medium
☐	710	1334	Find the City With the Smallest Number of Neighbors at a Threshold Distance	Medium
☐	711	1129	Shortest Path with Alternating Colors	Medium
☐	712	2662	Minimum Cost of a Path With Special Roads	Medium
☐	713	1368	Minimum Cost to Make at Least One Valid Path in a Grid	Hard
☐	714	2290	Minimum Obstacle Removal to Reach Corner	Hard
☐	715	2577	Minimum Time to Visit a Cell In a Grid	Hard
☐	716	882	Reachable Nodes In Subdivided Graph	Hard

Minimum Spanning Tree - 3 problems

Idea: Kruskal (sort edges + DSU) or Prim (heap)

☐	717	1584	Min Cost to Connect All Points	Medium
☐	718	GFG	Connecting Cities With Minimum Cost	Medium
☐	719	1489	Find Critical and Pseudo-Critical Edges in Minimum Spanning Tree	Hard

Advanced Graph Algorithms - 9 problems

Idea: Tarjan low-link for bridges/SCC; Hierholzer for Euler paths

☐	720	851	Loud and Rich	Medium
☐	721	1192	Critical Connections in a Network	Hard
☐	722	332	Reconstruct Itinerary	Hard
☐	723	753	Cracking the Safe	Hard
☐	724	2127	Maximum Employees to Be Invited to a Meeting	Hard
☐	725	GFG	Kosaraju's Strongly Connected Components	Hard
☐	726	GFG	Articulation Points in a Graph	Hard
☐	727	2360	Longest Cycle in a Graph	Hard
☐	728	2642	Design Graph With Shortest Path Calculator	Hard

Dynamic Programming (144 problems)

Ti ck	Sr	LC #	Problem	Level
1D DP - Fibonacci Style - 20 problems				
<i>Idea: $dp[i][j]$ depends on a few previous states; roll variables to $O(1)$ space</i>				
☐	729	70	Climbing Stairs	Easy
☐	730	1137	N-th Tribonacci Number	Easy
☐	731	746	Min Cost Climbing Stairs	Easy
☐	732	198	House Robber	Medium
☐	733	213	House Robber II	Medium
☐	734	740	Delete and Earn	Medium
☐	735	91	Decode Ways	Medium
☐	736	55	Jump Game	Medium
☐	737	45	Jump Game II	Medium
☐	738	139	Word Break	Medium
☐	739	790	Domino and Tromino Tiling	Medium
☐	740	1641	Count Sorted Vowel Strings	Medium
☐	741	GFG	Paint Fence	Medium
☐	742	935	Knight Dialer	Medium
☐	743	2140	Solving Questions With Brainpower	Medium
☐	744	2466	Count Ways To Build Good Strings	Medium
☐	745	1911	Maximum Alternating Subsequence Sum	Medium
☐	746	403	Frog Jump	Hard
☐	747	1340	Jump Game V	Hard
☐	748	1553	Minimum Number of Days to Eat N Oranges	Hard
0/1 Knapsack - 12 problems				
<i>Idea: Each item used once; iterate capacity backwards in 1D</i>				
☐	749	416	Partition Equal Subset Sum	Medium
☐	750	494	Target Sum	Medium
☐	751	1049	Last Stone Weight II	Medium
☐	752	474	Ones and Zeroes	Medium
☐	753	GFG	0/1 Knapsack Problem	Medium
☐	754	GFG	Subset Sum Problem	Medium
☐	755	GFG	Count of Subsets with Given Difference	Medium
☐	756	1155	Number of Dice Rolls With Target Sum	Medium
☐	757	879	Profitable Schemes	Hard
☐	758	956	Tallest Billboard	Hard
☐	759	2218	Maximum Value of K Coins From Piles	Hard
☐	760	2035	Partition Array Into Two Arrays to Minimize Sum Difference	Hard

Unbounded Knapsack - 7 problems

Idea: Items reusable; iterate capacity forwards

☐	761	322	Coin Change	Medium
☐	762	518	Coin Change II	Medium
☐	763	279	Perfect Squares	Medium
☐	764	377	Combination Sum IV	Medium
☐	765	343	Integer Break	Medium
☐	766	983	Minimum Cost For Tickets	Medium
☐	767	GFG	Rod Cutting Problem	Medium

Grid / Path DP - 18 problems

Idea: $dp[i][j]$ from top and left (and diagonal)

☐	768	62	Unique Paths	Medium
☐	769	63	Unique Paths II	Medium
☐	770	64	Minimum Path Sum	Medium
☐	771	120	Triangle	Medium
☐	772	931	Minimum Falling Path Sum	Medium
☐	773	221	Maximal Square	Medium
☐	774	1277	Count Square Submatrices with All Ones	Medium
☐	775	576	Out of Boundary Paths	Medium
☐	776	688	Knight Probability in Chessboard	Medium
☐	777	GFG	Longest Line of Consecutive One in Matrix	Medium
☐	778	1937	Maximum Number of Points with Cost	Medium
☐	779	2304	Minimum Path Cost in a Grid	Medium
☐	780	2400	Number of Ways to Reach a Position After Exactly k Steps	Medium
☐	781	174	Dungeon Game	Hard
☐	782	741	Cherry Pickup	Hard
☐	783	1463	Cherry Pickup II	Hard
☐	784	1301	Number of Paths with Max Score	Hard
☐	785	2435	Paths in Matrix Whose Sum Is Divisible by K	Hard

Longest Increasing Subsequence Family - 15 problems

Idea: $O(n^2)$ DP, or patience sorting with binary search in $O(n \log n)$

☐	786	674	Longest Continuous Increasing Subsequence	Easy
☐	787	300	Longest Increasing Subsequence	Medium
☐	788	673	Number of Longest Increasing Subsequence	Medium
☐	789	368	Largest Divisible Subset	Medium
☐	790	646	Maximum Length of Pair Chain	Medium
☐	791	1048	Longest String Chain	Medium
☐	792	1027	Longest Arithmetic Subsequence	Medium
☐	793	1218	Longest Arithmetic Subsequence of Given Difference	Medium
☐	794	GFG	Maximum Sum Increasing Subsequence	Medium
☐	795	GFG	Longest Bitonic Subsequence	Medium
☐	796	1626	Best Team With No Conflicts	Medium
☐	797	354	Russian Doll Envelopes	Hard
☐	798	1671	Minimum Number of Removals to Make Mountain Array	Hard
☐	799	1187	Make Array Strictly Increasing	Hard
☐	800	1691	Maximum Height by Stacking Cuboids	Hard

Two-String DP (LCS / Edit Distance) - 16 problems

Idea: $dp[i][j]$ over prefixes of both strings

☐	801	1143	Longest Common Subsequence	Medium
☐	802	583	Delete Operation for Two Strings	Medium
☐	803	712	Minimum ASCII Delete Sum for Two Strings	Medium
☐	804	1035	Uncrossed Lines	Medium
☐	805	718	Maximum Length of Repeated Subarray	Medium
☐	806	72	Edit Distance	Medium
☐	807	97	Interleaving String	Medium
☐	808	516	Longest Palindromic Subsequence	Medium
☐	809	115	Distinct Subsequences	Hard
☐	810	1092	Shortest Common Supersequence	Hard
☐	811	1312	Minimum Insertion Steps to Make a String Palindrome	Hard
☐	812	132	Palindrome Partitioning II	Hard
☐	813	10	Regular Expression Matching	Hard
☐	814	44	Wildcard Matching	Hard
☐	815	730	Count Different Palindromic Subsequences	Hard
☐	816	87	Scramble String	Hard

Interval DP - 15 problems*Idea: $dp[l][r]$ built from smaller intervals; try every split point k*

☐	817	486	Predict the Winner	Medium
☐	818	877	Stone Game	Medium
☐	819	1140	Stone Game II	Medium
☐	820	375	Guess Number Higher or Lower II	Medium
☐	821	1039	Minimum Score Triangulation of Polygon	Medium
☐	822	1690	Stone Game VII	Medium
☐	823	1130	Minimum Cost Tree From Leaf Values	Medium
☐	824	1406	Stone Game III	Hard
☐	825	312	Burst Balloons	Hard
☐	826	1547	Minimum Cost to Cut a Stick	Hard
☐	827	664	Strange Printer	Hard
☐	828	546	Remove Boxes	Hard
☐	829	GFG	Matrix Chain Multiplication	Hard
☐	830	1745	Palindrome Partitioning IV	Hard
☐	831	1000	Minimum Cost to Merge Stones	Hard

Stock Problems (State Machine DP) - 4 problems*Idea: States: hold / not hold / cooldown; transitions per day*

☐	832	309	Best Time to Buy and Sell Stock with Cooldown	Medium
☐	833	714	Best Time to Buy and Sell Stock with Transaction Fee	Medium
☐	834	123	Best Time to Buy and Sell Stock III	Hard
☐	835	188	Best Time to Buy and Sell Stock IV	Hard

Bitmask DP - 10 problems*Idea: mask = set of used items; $dp[mask][last]$*

☐	836	464	Can I Win	Medium
☐	837	1986	Minimum Number of Work Sessions to Finish the Tasks	Medium
☐	838	1947	Maximum Compatibility Score Sum	Medium
☐	839	1125	Smallest Sufficient Team	Hard
☐	840	943	Find the Shortest Superstring	Hard
☐	841	1349	Maximum Students Taking Exam	Hard
☐	842	1434	Number of Ways to Wear Different Hats to Each Other	Hard
☐	843	1879	Minimum XOR Sum of Two Arrays	Hard
☐	844	GFG	Travelling Salesman Problem	Hard
☐	845	1655	Distribute Repeating Integers	Hard

Digit DP - 4 problems*Idea: Build numbers digit by digit with (pos, tight, state) memo*

☐	846	902	Numbers At Most N Given Digit Set	Hard
☐	847	233	Number of Digit One	Hard
☐	848	600	Non-negative Integers without Consecutive Ones	Hard
☐	849	1012	Numbers With Repeated Digits	Hard

Counting and Partition DP - 23 problems

Idea: Count ways; often $dp[i][k]$ = split first i items into k groups

☐	850	413	Arithmetic Slices	Medium
☐	851	1043	Partition Array for Maximum Sum	Medium
☐	852	1395	Count Number of Teams	Medium
☐	853	1105	Filling Bookcase Shelves	Medium
☐	854	2370	Longest Ideal Subsequence	Medium
☐	855	808	Soup Servings	Medium
☐	856	GFG	Tree Diameter	Medium
☐	857	2008	Maximum Earnings From Taxi	Medium
☐	858	1223	Dice Roll Simulation	Medium
☐	859	446	Arithmetic Slices II - Subsequence	Hard
☐	860	1220	Count Vowels Permutation	Hard
☐	861	552	Student Attendance Record II	Hard
☐	862	920	Number of Music Playlists	Hard
☐	863	689	Maximum Sum of 3 Non-Overlapping Subarrays	Hard
☐	864	1278	Palindrome Partitioning III	Hard
☐	865	887	Super Egg Drop	Hard
☐	866	1639	Number of Ways to Form a Target String Given a Dictionary	Hard
☐	867	1420	Build Array Where You Can Find The Maximum Exactly K Comparisons	Hard
☐	868	1335	Minimum Difficulty of a Job Schedule	Hard
☐	869	1235	Maximum Profit in Job Scheduling	Hard
☐	870	1269	Number of Ways to Stay in the Same Place After Some Steps	Hard
☐	871	1359	Count All Valid Pickup and Delivery Options	Hard
☐	872	1416	Restore The Array	Hard

Greedy (47 problems)

Ti ck	Sr	LC #	Problem	Level
Sort + Local Best Choice - 21 problems				
<i>Idea: Sort by the right key, then take the best remaining option</i>				
☐	873	2144	Minimum Cost of Buying Candies With Discount	Easy
☐	874	1710	Maximum Units on a Truck	Easy
☐	875	860	Lemonade Change	Easy
☐	876	2037	Minimum Number of Moves to Seat Everyone	Easy
☐	877	2335	Minimum Amount of Time to Fill Cups	Easy
☐	878	2389	Longest Subsequence With Limited Sum	Easy
☐	879	1217	Minimum Cost to Move Chips to The Same Position	Easy
☐	880	406	Queue Reconstruction by Height	Medium
☐	881	1029	Two City Scheduling	Medium
☐	882	1833	Maximum Ice Cream Bars	Medium
☐	883	870	Advantage Shuffle	Medium
☐	884	846	Hand of Straights	Medium
☐	885	2279	Maximum Bags With Full Capacity of Rocks	Medium
☐	886	1561	Maximum Number of Coins You Can Get	Medium
☐	887	1921	Eliminate Maximum Number of Monsters	Medium
☐	888	539	Minimum Time Difference	Medium
☐	889	1551	Minimum Operations to Make Array Equal	Medium
☐	890	453	Minimum Moves to Equal Array Elements	Medium
☐	891	462	Minimum Moves to Equal Array Elements II	Medium
☐	892	1846	Maximum Element After Decreasing and Rearranging	Medium
☐	893	1402	Reducing Dishes	Hard
Jump, Reach and Circular Greedy - 4 problems				
<i>Idea: Track the farthest reach or a running surplus</i>				
☐	894	134	Gas Station	Medium
☐	895	1871	Jump Game VII	Medium
☐	896	1326	Minimum Number of Taps to Open to Water a Garden	Hard
☐	897	135	Candy	Hard

Greedy on Strings and Digits - 22 problems

Idea: Fix the most significant position first

☐	898	1221	Split a String in Balanced Strings	Easy
☐	899	409	Longest Palindrome	Easy
☐	900	1323	Maximum 69 Number	Easy
☐	901	944	Delete Columns to Make Sorted	Easy
☐	902	763	Partition Labels	Medium
☐	903	738	Monotone Increasing Digits	Medium
☐	904	670	Maximum Swap	Medium
☐	905	1663	Smallest String With A Given Numeric Value	Medium
☐	906	1647	Minimum Deletions to Make Character Frequencies Unique	Medium
☐	907	1653	Minimum Deletions to Make String Balanced	Medium
☐	908	1864	Minimum Number of Swaps to Make the Binary String Alternating	Medium
☐	909	376	Wiggle Subsequence	Medium
☐	910	1007	Minimum Domino Rotations For Equal Row	Medium
☐	911	1296	Divide Array in Sets of K Consecutive Numbers	Medium
☐	912	2870	Minimum Number of Operations to Make Array Empty	Medium
☐	913	861	Score After Flipping Matrix	Medium
☐	914	991	Broken Calculator	Medium
☐	915	1414	Find Minimum Number of Fibonacci Numbers Whose Sum Is K	Medium
☐	916	1717	Maximum Score From Removing Substrings	Medium
☐	917	330	Patching Array	Hard
☐	918	321	Create Maximum Number	Hard
☐	919	2193	Minimum Number of Moves to Make Palindrome	Hard

Bit Manipulation (28 problems)

Ti ck	Sr	LC #	Problem	Level
Bit Counting and Tricks - 14 problems				
<i>Idea: $n \& (n-1)$ clears the lowest set bit; $n \& -n$ isolates it</i>				
☐	920	191	Number of 1 Bits	Easy
☐	921	338	Counting Bits	Easy
☐	922	190	Reverse Bits	Easy
☐	923	342	Power of Four	Easy
☐	924	461	Hamming Distance	Easy
☐	925	476	Number Complement	Easy
☐	926	1009	Complement of Base 10 Integer	Easy
☐	927	693	Binary Number with Alternating Bits	Easy
☐	928	405	Convert a Number to Hexadecimal	Easy
☐	929	477	Total Hamming Distance	Medium
☐	930	201	Bitwise AND of Numbers Range	Medium
☐	931	393	UTF-8 Validation	Medium
☐	932	1318	Minimum Flips to Make a OR b Equal to c	Medium
☐	933	1611	Minimum One Bit Operations to Make Integers Zero	Hard
XOR Properties - 9 problems				
<i>Idea: $x \wedge x = 0$ and $x \wedge 0 = x$; XOR of a range via prefix XOR</i>				
☐	934	389	Find the Difference	Easy
☐	935	1720	Decode XORed Array	Easy
☐	936	2433	Find the Original Array of Prefix Xor	Easy
☐	937	137	Single Number II	Medium
☐	938	260	Single Number III	Medium
☐	939	1734	Decode XORed Permutation	Medium
☐	940	1310	XOR Queries of a Subarray	Medium
☐	941	1442	Count Triplets That Can Form Two Arrays of Equal XOR	Medium
☐	942	1835	Find XOR Sum of All Pairs Bitwise AND	Hard
Bit Arithmetic and Masks - 5 problems				
<i>Idea: Emulate + - / with shifts; use masks as tiny sets</i>				
☐	943	371	Sum of Two Integers	Medium
☐	944	29	Divide Two Integers	Medium
☐	945	318	Maximum Product of Word Lengths	Medium
☐	946	898	Bitwise ORs of Subarrays	Medium
☐	947	2419	Longest Subarray With Maximum Bitwise AND	Medium

Math and Number Theory (33 problems)

Ti ck	Sr	LC #	Problem	Level
Digits and Simple Arithmetic - 14 problems				
<i>Idea: Use % 10 and / 10; watch integer overflow</i>				
☐	948	412	Fizz Buzz	Easy
☐	949	258	Add Digits	Easy
☐	950	728	Self Dividing Numbers	Easy
☐	951	1025	Divisor Game	Easy
☐	952	1103	Distribute Candies to People	Easy
☐	953	2224	Minimum Number of Operations to Convert Time	Easy
☐	954	1266	Minimum Time Visiting All Points	Easy
☐	955	7	Reverse Integer	Medium
☐	956	400	Nth Digit	Medium
☐	957	166	Fraction to Recurring Decimal	Medium
☐	958	2579	Count Total Number of Colored Cells	Medium
☐	959	397	Integer Replacement	Medium
☐	960	319	Bulb Switcher	Medium
☐	961	273	Integer to English Words	Hard
Primes, Powers and GCD - 12 problems				
<i>Idea: Sieve for primes; Euclid for gcd; fast exponentiation</i>				
☐	962	326	Power of Three	Easy
☐	963	263	Ugly Number	Easy
☐	964	1071	Greatest Common Divisor of Strings	Easy
☐	965	204	Count Primes	Medium
☐	966	172	Factorial Trailing Zeroes	Medium
☐	967	365	Water and Jug Problem	Medium
☐	968	372	Super Pow	Medium
☐	969	866	Prime Palindrome	Medium
☐	970	1998	GCD Sort of an Array	Hard
☐	971	1819	Number of Different Subsequences GCDs	Hard
☐	972	483	Smallest Good Base	Hard
☐	973	479	Largest Palindrome Product	Hard
Geometry and Puzzles - 7 problems				
<i>Idea: Reduce to cross products, slopes (via gcd) or invariants</i>				
☐	974	1344	Angle Between Hands of a Clock	Medium
☐	975	1041	Robot Bounded In Circle	Medium
☐	976	223	Rectangle Area	Medium
☐	977	593	Valid Square	Medium
☐	978	149	Max Points on a Line	Hard
☐	979	391	Perfect Rectangle	Hard
☐	980	458	Poor Pigs	Hard

Sorting and Advanced DS (20 problems)

Ti ck	Sr	LC #	Problem	Level
Sorting Algorithms and Applications - 9 problems <i>Idea: Know merge sort (counts inversions) and quickselect cold</i>				
☐	981	1122	Relative Sort Array	Easy
☐	982	1636	Sort Array by Increasing Frequency	Easy
☐	983	912	Sort an Array	Medium
☐	984	274	H-Index	Medium
☐	985	GFG	Merge Sort	Medium
☐	986	GFG	Quick Sort	Medium
☐	987	GFG	Count Inversions	Medium
☐	988	164	Maximum Gap	Hard
☐	989	493	Reverse Pairs	Hard
Fenwick Tree (BIT) and Segment Tree - 11 problems <i>Idea: Point update + range query in $O(\log n)$</i>				
☐	990	307	Range Sum Query - Mutable	Medium
☐	991	1828	Queries on Number of Points Inside a Circle	Medium
☐	992	GFG	Range Minimum Query	Medium
☐	993	315	Count of Smaller Numbers After Self	Hard
☐	994	327	Count of Range Sum	Hard
☐	995	699	Falling Squares	Hard
☐	996	218	The Skyline Problem	Hard
☐	997	2407	Longest Increasing Subsequence II	Hard
☐	998	2179	Count Good Triplets in an Array	Hard
☐	999	GFG	Range Sum Query 2D - Mutable	Hard
☐	1000	GFG	Sparse Table for Range Queries	Hard

Quick Pattern Checklist

Revision reminders - when you see these signals, reach for the matching pattern:

If you see	Try
Sorted array / pair with target	Two pointers or binary search
Subarray / substring, contiguous	Sliding window or prefix sum + hash map
"Minimum/maximum x such that feasible"	Binary search on the answer
Next greater / smaller element	Monotonic stack
Top-K / k-th largest / streaming median	Heap (two heaps for median)
All combinations / permutations / placements	Backtracking
Shortest path, unweighted	BFS (multi-source if many starts)
Dependencies / ordering	Topological sort
Connected components / cycles in undirected graph	Union Find
Weighted shortest path	Dijkstra (Bellman-Ford if $\leq K$ edges or negatives)
Overlapping subproblems, count ways / min cost	Dynamic programming
Prefix lookup / autocomplete	Trie
Linked list cycle or middle	Fast and slow pointers
Locally best choice provably safe	Greedy (sort first)